

## Week 6

# Introduction to Game Theory

Mon 13 Apr 26 - Sun 19 Apr 26

### Overview

We now switch away from reinforcement learning and look at Game Theory, a mathematical theory of strategic interaction (you will find out what this means :). This is arguably the most important mathematical framework for analysing multi-agent interactions. We start with classical non-cooperative game theory.

## Your Learning Journey

### Own-time



### Real-time



### Wrap-up



### Own-time

## Your schedule for the week

| Tasks                                                                                    | Complete By/Before | Time Expected     |
|------------------------------------------------------------------------------------------|--------------------|-------------------|
| Listen to the mini podcast [ experimental ]                                              | Before the seminar | 15 minutes        |
| Participate in the seminar & tutorial                                                    | End of the week    | 2 hours + 2 hours |
| Complete the essential tutorial components if you didn't manage this during the tutorial | End of the week    | [ individual ]    |
| Consolidate your learning with the extended podcast and chat space [ experimental ]      | End of the Week    | 1 hour            |

[Mini-Podcast: Introduction to Game Theory](#) M4A

AI-generated mini podcast setting the scene for the introduction to temporal difference learning (1'50')

The AI-generated podcasts are generated from resources curated by me and vetted before release. However, they are still not fully authoritative resources and you should not blindly trust them. In fact, part of our experiment with the podcasts is that you are meant to come back with questions about what is said in the podcasts that we can then discuss in the seminar and in the tutorials. In particular, for these discussions you should try to identify elements that you think might be incorrect!

2.8 MB

## Real-time



### Seminar Resources

 [Week 6 - Introduction to Game Theory](#) PDF

3.8 MB

### Lab Resources

 [Lab Week 6 Instructions](#) PDF

148.0 KB

 [Deep-Q Skeleton code](#) PY

2.5 KB

 [Deep Q Skeleton API Use Example \(Jupyter Notebook\)](#) IPYNB

10.2 KB

 [Sample Code for Tabular Q-Learning Problem \(Python\)](#) IPYNB

36.5 KB

 [Sample Code for Tabular Q-Learning Problem \(MMA\)](#) NB

277.4 KB

### Solutions to selected exercises

 [Gridworld Library](#) PY

9.3 KB

 [Deep Q Example Solution \(Jupyter Notebook\)](#) IPYNB

211.9 KB



Week 6 Lab Zoom Link for Suzhou



Not available unless: You belong to **CALLISTA: FIT5226 OS-CHI-SEU ON-CAMPUS S1 2026**



## Wrap-up



[AI Review Podcast \[ experimental \]: Introduction to Non-cooperative Game Theory](#) M4A

39.7 MB



[AI Chat Space \[ experimental \]: Introduction to Non-cooperative Game Theory](#)

Please keep in mind that these experimental AI chat spaces are not authoritative sources. While they work from curated authoritative sources they can and will make mistakes. Use them for guidance but make sure that you always check everything using the authoritative literature sources.